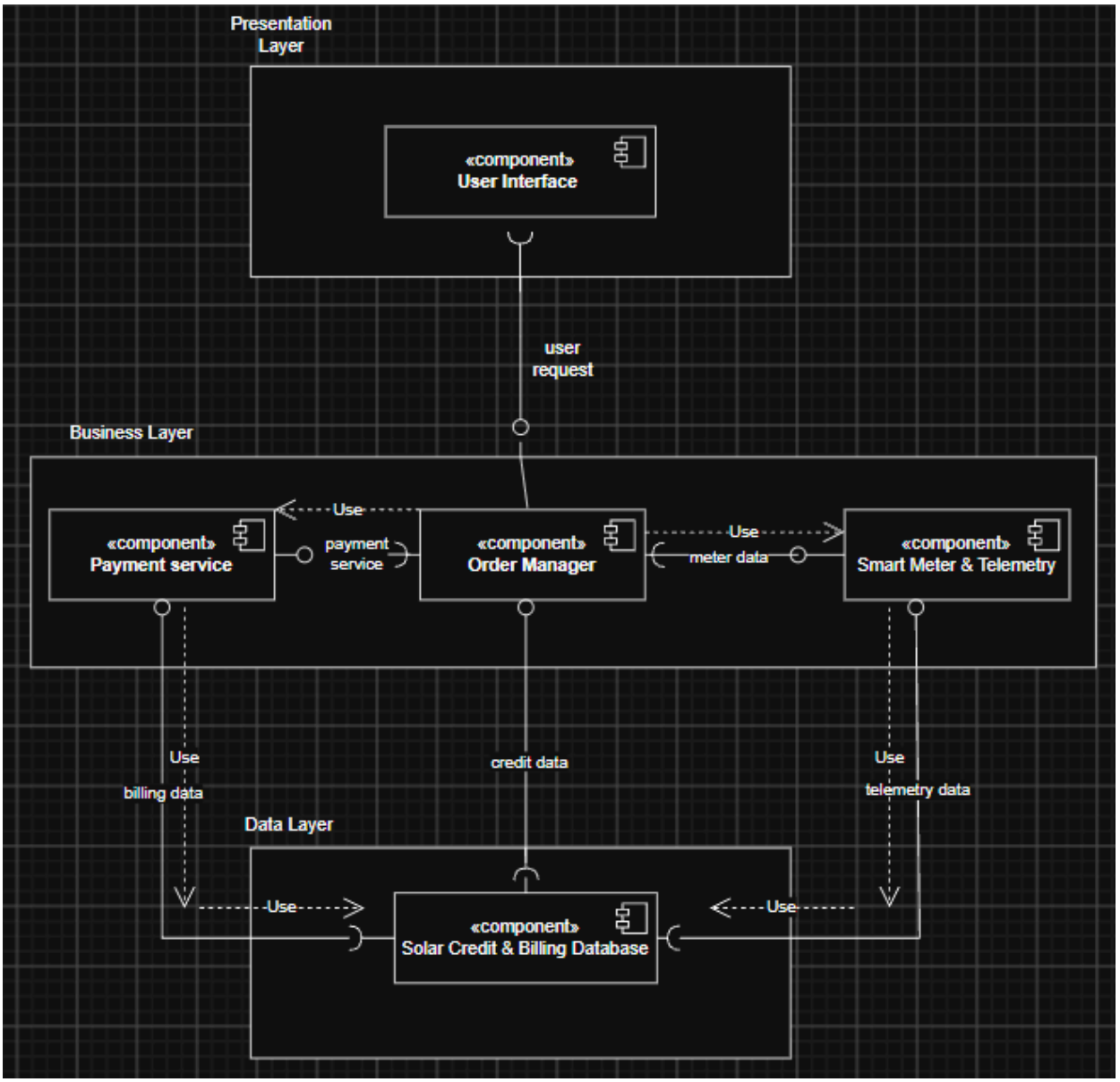
SE BANANA PROBLEM

LAB - 3

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UML Component diagram -



Architecture Selection - Community Solar Credit Allocation Manager

I chose Layered Architecture for the Community Solar Credit Allocation Manager.

Architectural Choice

The system uses a Layered Architecture consisting of Presentation, Business, and Data layers. The Presentation Layer contains the User Interface, the Business Layer contains the Order Manager, Payment Service, and Smart Meter & Telemetry components, and the Data Layer contains the Solar Credit & Billing Database.

Reason 1 - Separation of Responsibilities

Layered Architecture separates user interaction, business processing, and data management. This is suitable for the system because credit transfers, billing-offset processing, and smart-meter telemetry can be handled in the Business Layer without directly exposing database operations to users.

Reason 2 - Centralized Credit and Billing Data

The system needs to maintain solar-generation records, credit balances, transactions, and monthly billing offsets. A dedicated Data Layer provides centralized storage and makes these records easier to manage consistently.

Security Advantage

Security controls can be applied at different layers. User access can be authenticated at the Presentation Layer, while the Business Layer can validate credit-transfer requests and prevent transfers that exceed the available credit balance before updating the database.

Performance Benefit

The Smart Meter & Telemetry component can validate hourly generation data before storing it in the database. Separating telemetry processing from the user interface helps the system handle generation data for the required 500 households while supporting the 99.9% uptime requirement.